

Scaling Properties of Diffusion Models for Perceptual Tasks

RAHUL RAVISHANKAR*, ZEESHAN PATEL*, JATHUSHAN RAJASEGARAN, and JITENDRA MALIK, University of California, Berkeley, USA

In this paper, we argue that iterative computation with diffusion models offers a powerful paradigm for not only generation but also visual perception tasks. We unify tasks such as depth estimation, optical flow, and amodal segmentation under the framework of image-to-image translation, and show how diffusion models benefit from scaling training and test-time compute for these perceptual tasks. Through a careful analysis of these scaling properties, we formulate compute optimal training and inference recipes to scale diffusion models for visual perception tasks. Our models achieve competitive performance to state-of-the-art methods using significantly less data and compute. We release code and models at [scaling-diffusion-perception.github.io](https://github.com/ravishankar/scaling-diffusion-perception).

ACM Reference Format:

Rahul Ravishankar, Zeeshan Patel, Jathushan Rajasegaran, and Jitendra Malik. 2026. Scaling Properties of Diffusion Models for Perceptual Tasks. 1, 1 (July 2026), 14 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Diffusion models have become a leading approach for generating images and video, and they scale remarkably well as data, model size, and compute increase. Here, we introduce a single, unified diffusion model capable of handling several distinct perceptual tasks at once — depth estimation, optical flow estimation, and amodal segmentation — illustrated in Figure 1.

A number of prior efforts, including Marigold [21], FlowDiffuser [29], and pix2gestalt [32], have already shown that pre-trained image diffusion models can be adapted to individual inverse-vision problems. We build on this line of work with a broad empirical study: we derive scaling power laws for depth estimation and show that these laws carry over to the other perceptual tasks as well. These scaling trends then inform compute-optimal recipes for both training and inference. Overall, we find that scaling compute efficiently yields substantial performance gains across all three tasks, without requiring the massive pre-training budgets used by prior methods.

A parallel trend in language modeling has been to scale test-time, rather than training-time, compute to boost reasoning ability, as seen in OpenAI’s o1 model [31]. Noam Brown, one of the researchers behind that work, captured the idea memorably in a talk: letting a poker-playing agent think for a mere 20 extra seconds per hand matched the gains from scaling the underlying model up by a factor of 100,000 and training it 100,000 times longer. We observe an analogous trade-off in our own experiments: allocating compute at test time versus training time both move the needle on downstream perceptual-task performance, and the two axes can be balanced against one another.

*Equal Contribution

Authors’ Contact Information: Rahul Ravishankar, rravishankar@berkeley.edu; Zeeshan Patel, zeeshanp@berkeley.edu; Jathushan Rajasegaran; Jitendra Malik, University of California, Berkeley, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

We exploit diffusion’s iterative, stochastic sampling process to scale test-time compute directly, primarily by increasing the number of denoising steps. Spending more of that additional compute on the earliest denoising steps, and ensembling several denoised predictions together, consistently improves accuracy across our perceptual tasks. Taken together, our results show that scaling test-time compute is a practical lever for inverse-vision problems operating under tight compute budgets, offering an alternative to the usual emphasis on scaling training compute alone.

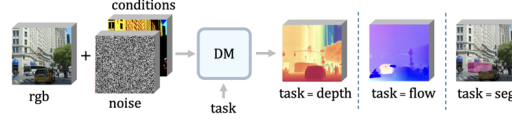


Fig. 1. **A Unified Framework:** We fine-tune a pre-trained Diffusion Model (DM), for visual perception tasks. We take a RGB image, and a conditional image (i.e. next video frame, occlusion mask, etc.), along with the noised image of the ground truth prediction. Our model generates predictions for visual tasks such as depth estimation, optical flow prediction, and amodal segmentation, based on the conditional task embedding. We train a generalist model that can perform all three tasks with exceptional performance.

2 Related Work

Generative Modeling: The generative modeling literature spans several major families of methods: VAEs [22], GANs [14], normalizing flows [35], autoregressive models [49], and diffusion models [17, 43]. Among these, Denoising Diffusion Probabilistic Models (DDPMs) [17] have proven especially good at scaling for image and video generation. Well-known examples include Latent Diffusion Models [36], which gain efficiency by operating in a compressed latent space rather than pixel space; Imagen [38], which instead generates directly in pixel space at progressively higher resolutions; and Consistency Models [45], which target faster sampling without sacrificing generation quality. More recent approaches such as Rectified Flow [28] and Flow Matching [26] borrow training objectives from optimal transport theory to learn continuous vector fields between data and target distributions, sidestepping the discrete-timestep formulation that standard diffusion relies on. Rectified Flow addresses training instabilities through flow regularization, while Flow Matching enables efficient sampling with fewer discretization artifacts, making both attractive alternatives to conventional diffusion. Outside of diffusion, Parti [55] and MARS [16] have demonstrated strong autoregressive image generation, and Muse [6] introduced a transformer-based masked-image-generation approach.

Scaling Diffusion Models: Diffusion models have also displayed strong scaling behavior along the axes of data, model size, and compute. Latent Diffusion Models [36] were among the first to show that training on large web-scale datasets with a U-Net backbone yields high-quality image generation. DiT [33] instead swapped in a transformer backbone, revealing favorable scaling properties for class-conditional generation. Building on this, Li et al. [25] examined alignment scaling laws for text-to-image diffusion, and Fei et al. [10] pushed mixture-of-experts DiT models up to 16B parameters while maintaining generation quality. A separate route to scaling transformers is upcycling: Komatsuzaki et al. [23] showed that a dense transformer language model can be converted into a mixture-of-experts model this way, without retraining from scratch. In a similar spirit, EC-DiT [46] studies heterogeneous compute allocation for mixture-of-experts DiT training, using expert-choice routing to adaptively assign compute to individual text-image samples.

Diffusion Models for Perception Tasks: Beyond generation, diffusion models have been applied to a range of downstream visual tasks, including depth estimation [8, 20, 39, 40, 56]. Marigold [21] and GeoWizard [13] are two more

recent examples, both repurposing pre-trained diffusion models for monocular depth estimation with strong results. With modest architectural changes, diffusion models have also been applied to semantic segmentation over categorical distributions [1–3, 18, 47, 53], instance segmentation [15], and panoptic segmentation [7]. Other applications include optical flow [29, 40] and 3D understanding [19, 27, 34, 50, 51].

3 Generative Pre-training

We begin by studying how to scale diffusion pre-training efficiently. Our diffusion models are trained for class-conditional image generation on a diffusion transformer (DiT) backbone, following the original DiT training recipe [33].

Starting with a target RGB image $I \in R^{uu3}$, where the resolution of the image is uu , our pretrained, frozen Stable Diffusion variational autoencoder [36] compresses the target to a latent $z_0 \in R^{ww4}$, where $w = u/8$. Gaussian noise is added at sampled time steps to obtain a noisy target latent. Noisy samples are generated as:

$$z_t = \sqrt{\alpha_t} \cdot z_0 + \sqrt{1 - \alpha_t} \cdot \epsilon_t \quad (1)$$

for timestep t . The noise is distributed as $\epsilon \sim \mathcal{N}(0, I)$, $t \sim \text{Uniform}(T)$, with $T = 1000$ and $\alpha_t := \prod_{s=1}^t (1 - \beta_s)$, with $\{\beta_1, \dots, \beta_T\}$ as the variance schedule of a process.

In the denoising process, the class-conditional DiT $f(\cdot)$, parameterized by learned parameters θ , gradually removes noise from z_t to obtain z_{t1} . The parameters θ are updated by noising z_0 with sampled noise ϵ at random time steps t , computing the noise estimate, and optimizing $\arg \min_{\theta} \mathcal{L}_{\theta}(z_t, \epsilon_i) = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n (\epsilon_i - \hat{\epsilon}_i)^2$ where θ are the DiT learned parameters and $\hat{\epsilon}_i$ is the DiT noise prediction for sample i .

3.1 Model Size

To study how pre-training scale affects performance, we train six dense DiT configurations of increasing size, summarized in Table 1, by varying both the number of transformer layers and the hidden dimension. All six are pre-trained on Imagenet-1K [37] for 400k iterations, using a fixed learning rate of $1e-4$ and a batch size of 256. As Fig. 2 shows, the bigger models consistently reach lower training loss, and the relationship follows a clean power law. Plotting train loss against compute (measured in MACs) yields the fitted relationship $L(C) = 0.23 \times C^{-0.0098}$. These pre-training results suggest that DiT scales gracefully even on a comparatively modest dataset, and this behavior carries over directly into stronger downstream performance after fine-tuning.

3.2 Mixture of Experts

Alongside the dense models, we pre-train Sparse Mixture-of-Experts (MoE) variants [41], adopting the $S/2$ and $L/2$ configurations from [11]. Table 2 lists the three MoE configurations we use, which scale total parameter count by increasing hidden size, expert count, layer depth, and attention heads. Every MoE block routes each token to its top-2 experts and additionally passes it through a shared expert used by all tokens. We adopt the expert-balance loss from [11] to keep load spread evenly across experts. This sparse MoE recipe lets us grow parameter count while also improving throughput, making it more compute-efficient than training a dense DiT model of comparable size. We train our DiT-MoE models with the same training recipe as the dense DiT model using ImageNet-1K. Our approach enables training DiT-MoE models to increase model capacity without increasing compute usage by another order of magnitude, which would be required to train dense models of similar sizes.

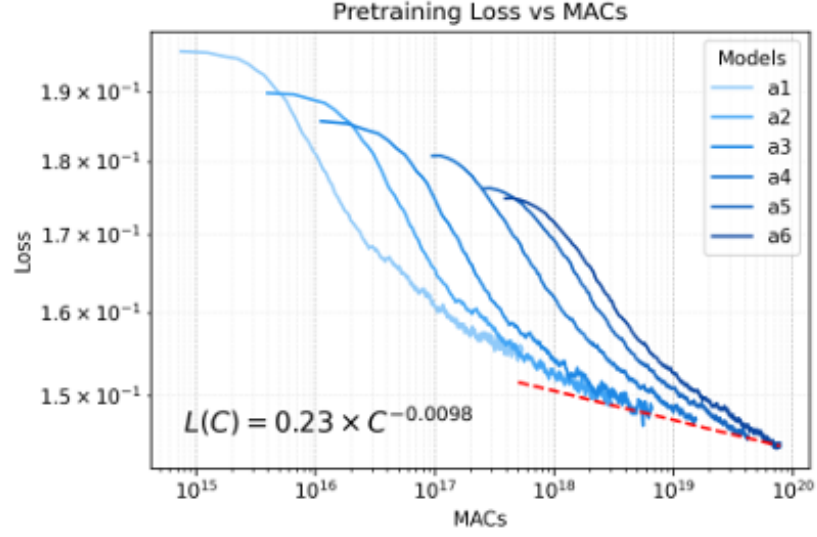


Fig. 2. **Scaling at Model Size:** For generative pre-training of DiT models, we observe clear power law scaling behavior as we increase the model size.

model	params	dimension	head	layers
a1	14.8M	256	16	12
a2	77.2M	512	16	16
a3	215M	768	16	20
a4	458M	1024	16	24
a5	1.2B	1536	16	28
a6	1.9B	1792	16	32

Table 1. **Dense DiT Models:** We scale dense DiT model size by increasing hidden dimension and number of layers linearly while keeping number of heads constant following [54].

Model	Active / Total	Dim	Heads	Layers
S/2-8E2A	71M / 199M	384	6	12
S/2-16E2A	71M / 369M	384	6	12
L/2-8E2A	1.0B / 2.8B	1024	16	24

Table 2. **MoE DiT Models:** We scale the MoE DiT models by increasing dimension size, number attention heads, layers, and experts following [11].

4 Fine-tuning for Perceptual Tasks

This section looks at how to scale fine-tuning of the pre-trained DiT models to get the most out of downstream perception performance. We adopt the image-to-image diffusion training recipe from [21] and [4], framing every visual task as conditional denoising diffusion generation. Given an RGB image $I \in R_{u \times u \times 3}$ and its paired ground-truth image $D \in R_{u \times u \times 3}$, we first project both into latent space, giving $i_0 \in R^{w \times w \times 4}$ and $d_0 \in R^{w \times w \times 4}$. Noise is added only to the ground-truth latent to produce d_t , which is then concatenated with the RGB latent to form $z_t = \{i_0, d_t\}$. To accommodate the doubled number of input channels, we modify the DiT model’s first convolutional layer and halve its weight values, so that predictions remain unchanged when the input is a plain RGB image, following [21]. Diffusion

training then proceeds by denoising toward the ground-truth image. We run our fine-tuning compute-scaling ablations on the monocular depth estimation task, reporting Absolute Relative error and Delta1 error, and later carry the best configurations over to the other perceptual tasks.

4.1 Effect of Model Size

To study the effect of model size on fine-tuning, we take the pre-trained a1–a6 dense models from Section 3.1 and fine-tune each on depth estimation. Fig. 3 shows that the larger DiT models reliably reach lower fine-tuning loss, again following a clear power law. Plotting both training loss and validation metrics against compute (in MACs), we find that fine-tuned model performance on both depth Absolute Relative error and depth Delta1 error follows the same power-law pattern. These results give a strong signal for how performance should scale as fine-tuning compute grows through model size alone.

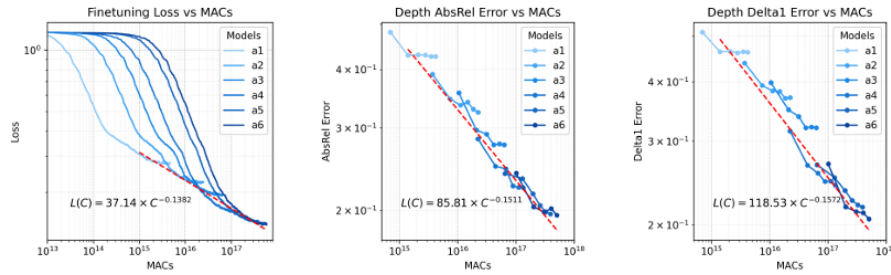


Fig. 3. **Effect of Model Size:** We fine-tune a1–a6 models on the Hypersim dataset for 30K iterations with an exponential decay learning rate schedule from $3e-5$ to $3e-7$. We observe a strong correlation between the fine-tuning loss scaling law and validation metric scaling laws.

4.2 Effect of Pre-training Compute

We next examine how fine-tuning behaves as we vary the amount of pre-training given to the DiT backbone. Four a4-configuration models are trained with different numbers of pre-training steps while every other hyperparameter is held fixed, and each is then fine-tuned on the same depth estimation dataset. Fig. 4 shows that the depth estimation validation metrics follow a power law as DiT pre-training steps increase, indicating that stronger pre-trained representations translate into more effective use of fine-tuning compute.

4.3 Effect of Image Resolution

Image resolution is another lever on training compute, since a higher-resolution input produces a longer token sequence for each forward pass. More tokens per image means more signal for the model to learn from at training time, which can strengthen its internal representations and, in turn, downstream performance. We take dense DiT-XL models pre-trained at 256×256 and 512×512 resolution from [33], along with DiT-MoE L/2-8E2A models pre-trained at the same two resolutions following [11], and fine-tune each at its native resolution for depth estimation. Figure 5 shows that scaling fine-tuning compute this way, through image resolution, produces meaningful gains in downstream depth estimation performance.

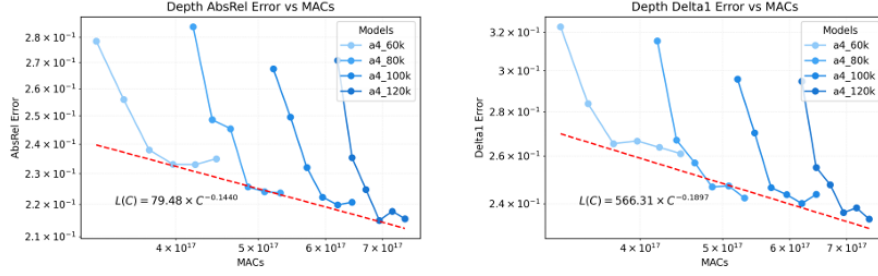


Fig. 4. **Effect of Scaling Model Pre-training Compute on Depth Estimation:** (a) Depth Absolute Relative Error vs. MACs. (b) Depth Delta1 Error vs. MACs. We pre-train four a4 models with 60K, 80K, 100K, and 120K steps. These models are then fine-tuned for 30K steps on the Hypersim depth estimation dataset. We observe a clear power law as we increase the DiT pre-training compute across depth estimation validation metrics.

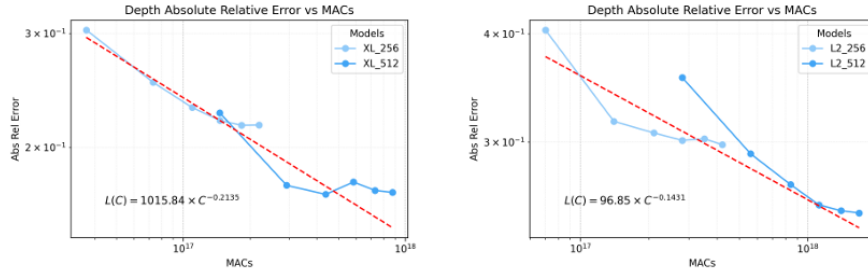


Fig. 5. **Effect of Image Resolution** We fine-tune DiT-XL and DiT-MoE L/2 models with resolutions of 256×256 and 512×512 . We observe a power law when increasing image resolution during training. By scaling the number of tokens per image by 4x, we achieve strong performance on Depth Absolute Error, displaying the effect of increasing total dataset tokens for dense visual perception tasks such as depth estimation.

4.4 Effect of Upcycling

While sparse MoE models offer an efficient way to grow model capacity, pre-training one from scratch is still costly. Sparse MoE Upcycling [24] offers a cheaper path: it turns a dense transformer checkpoint into an MoE model by duplicating each transformer block’s MLP layer E times (one per expert) and adding a learnable router that sends each token to its top- k experts, whose outputs are then combined in a weighted sum at the end of the block. We apply upcycling to several dense DiT models that have already been fine-tuned for depth estimation, and continue fine-tuning the resulting upcycled models. Fig. 6 shows the resulting scaling laws, with average gains of 5.3% on Absolute Relative Error and 8.6% on Delta1 error.

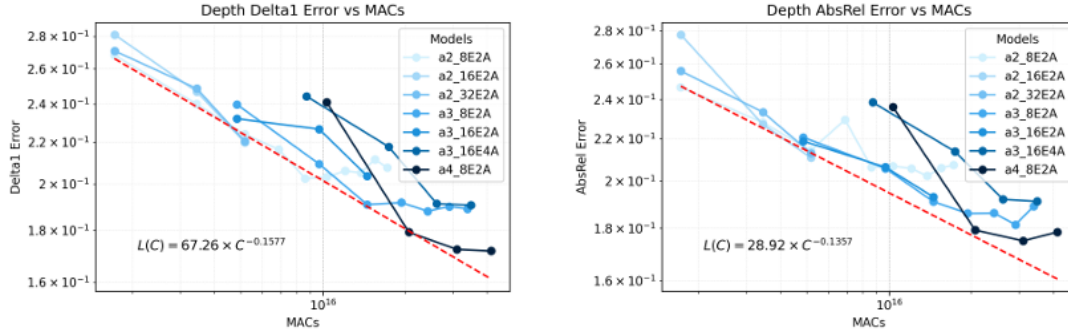


Fig. 6. **Effect of Upcycling.** We upcycle a2, a3, and a4 models fine-tuned for depth estimation with a varying number of total/active model experts. We continue fine-tuning each upcycled model for 15K iterations on the Hypersim depth estimation dataset. We observe a clear scaling law in the validation metrics as we increase fine-tuning compute with upcycling. The upcycled models can also achieve equivalent or superior performance to our dense a5 and a6 checkpoints, each of which utilize more compute during pre-training and fine-tuning. Increasing the total model experts and total active experts can also improve the downstream performance.

5 Scaling Test-time Compute

Scaling test-time compute is already a well-explored lever for autoregressive LLMs looking to improve long-horizon reasoning [5, 9, 31, 42]. Here, we show that the same lever reliably improves diffusion model performance on perceptual tasks. Fig. 7 summarizes our overall approach: we first encode the input image into latent space with the Stable-Diffusion VAE [36], then sample a target noise latent from a standard Gaussian and iteratively denoise it with DDIM [44] to obtain the final prediction.

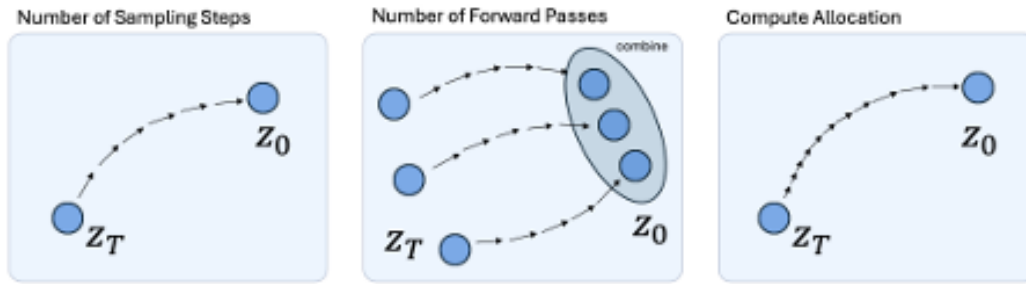


Fig. 7. **Inference Scaling:** Diffusion models by design allow efficient scaling of test-time compute. First, we can simply increase the number of denoising steps to increase the compute spent at inference. Since we are estimating deterministic outputs, we can then initialize multiple noise latents and ensemble the predictions to get a better estimation. Finally, we can also reallocate our test-time compute budget for low and high frequency denoising by modifying the noise variance schedule.

5.1 Effect of Scaling Inference Steps

The simplest way to scale diffusion inference is to add more denoising steps. Because the model is trained to denoise across a range of timesteps, increasing the number of steps at test time naturally yields finer, more accurate predictions.

This coarse-to-fine behavior, familiar from the generative setting, transfers directly to the discriminative case as well. Fig. 8 shows that simply increasing the number of diffusion sampling steps, and thereby the total test-time compute, produces substantial gains in depth estimation performance.

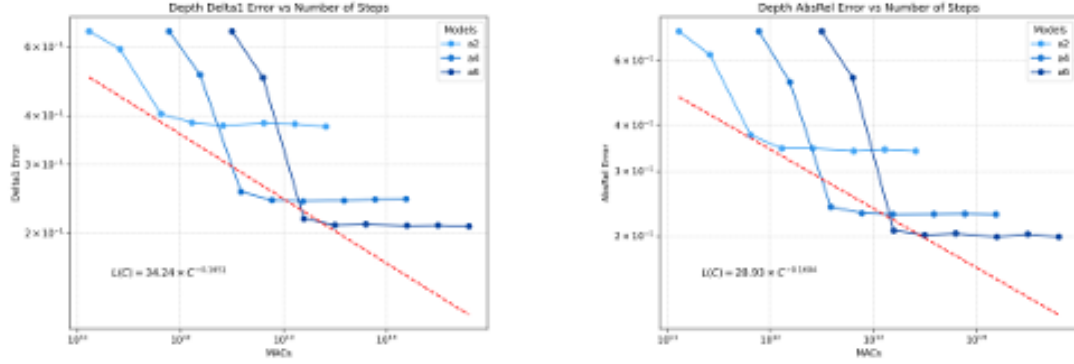


Fig. 8. **Effect of Number of Sampling Steps.** (a) Delta1 Error vs. Number of Steps. (b) Absolute Relative Error vs. Number of Steps. For each model, we sample for $T \in [1, 2, 5, 10, 20, 50, 100]$ steps with the DDIM sampler. We show a clear power law scaling behavior in (a) and (b), displaying the effectiveness of scaling test-time compute by increasing the number of diffusion sampling steps.

5.2 Effect of Test-time Ensembling

Test-time ensembling offers a second way to scale inference compute, exploiting the fact that denoising different noise latents yields different downstream predictions. Here, we run N forward passes per input sample and combine the outputs using one of two strategies. The first, naive ensembling, simply takes the pixel-wise median across all N outputs. The second, following Marigold [21], is median compilation, where we collect predictions $\{\hat{d}_1, \dots, \hat{d}_N\}$ that are affine-invariant, jointly estimate scale and shift parameters $\hat{s}_i$ and $\hat{t}_i$, and minimize the distances between each pair of scaled and shifted predictions $(\hat{d}'_i, \hat{d}'_j)$ where $\hat{d}' = \hat{d} \times \hat{s} + \hat{t}$. For each optimization step, we take the pixel-wise median $\mathbf{m}(x, y) = \text{median}(\hat{d}'_1(x, y), \dots, \hat{d}'_N(x, y))$ to compute the merged depth $\mathbf{m}$. Since it requires no ground truth, we scale ensembling by increasing N to utilize more test-time compute. Fig. 9 shows that this again produces a clear power-law improvement in depth estimation performance.

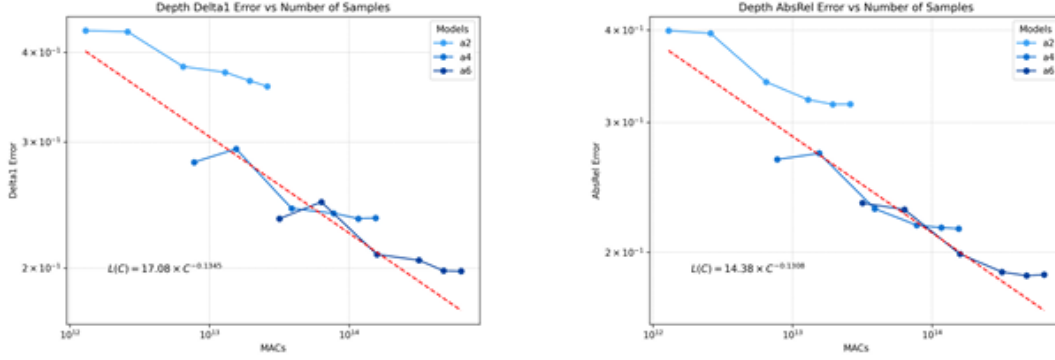


Fig. 9. **Effect of Test Time Ensembling.** (a) Delta1 Error vs. Number of Forward Passes. (b) Absolute Relative Error vs. Number of Forward Passes. Ensembling multiple predictions from distinct noise initializations displays power law scaling behavior. We apply test-time ensembling values of $N \in [1, 2, 5, 10, 15, 20]$.

5.3 Effect of Noise Variance Schedule

A third way to scale test-time compute is to change where, rather than how much, compute is spent during denoising. Diffusion noise schedulers define how the variance of the added Gaussian noise evolves across the T diffusion timesteps, and adjusting this schedule effectively reallocates compute toward earlier or later denoising steps. We compare three DDIM noise-level settings: linear, scaled linear, and cosine. The cosine schedule from [30] decreases linearly starting from the midpoint of the corruption process, avoiding the overly aggressive early corruption seen with linear schedules. Fig. 10 shows that, under a fixed compute budget, the cosine schedule outperforms linear schedules for DDIM on depth estimation.

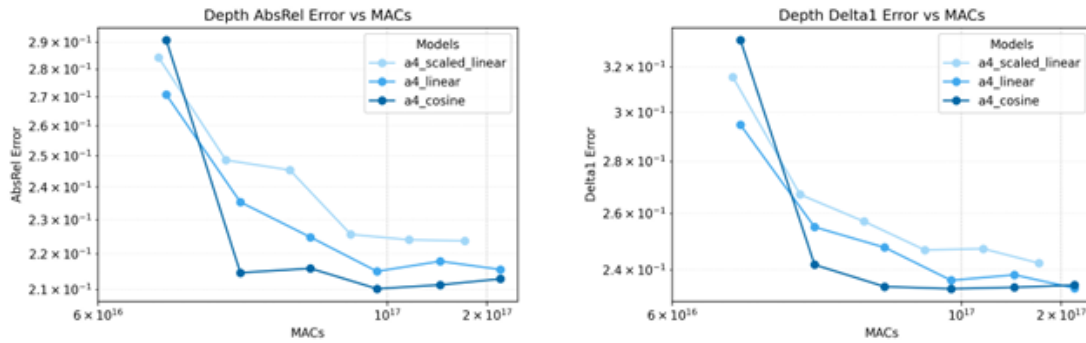


Fig. 10. **Effect of Noise Variance (Beta) Schedule.** We fine-tune a4 models with three different beta schedules: linear, scaled linear, cosine. Reallocating compute with the cosine schedule to spend more time denoising at earlier timesteps significantly improved Delta1 and Absolute Relative Error rates.

6 Putting It All Together

Building on the lessons from our depth estimation scaling experiments, we extend our approach to train diffusion models for optical flow prediction and amodal segmentation. Across all three tasks, our efficient training and test-time compute-scaling strategies deliver substantial performance gains, matching or exceeding current state-of-the-art methods. These experiments offer practical guidance for applying diffusion models to visual perception under constrained compute. We then go a step further and train a single unified model capable of all three tasks at once, demonstrating that our scaling techniques generalize well beyond any one task. Altogether, our results show that these training and test-time scaling strategies remove the dependence on internet-scale pre-training data for high-quality visual perception with diffusion models. Fig. 11 shows qualitative predictions from our models.

6.1 Depth Estimation

Combining what we learned from the depth estimation ablations, we assemble a model using the best training and inference settings identified so far. We fine-tune a DiT-XL model from [33] on Hypersim depth data for 30K steps, with a batch size of 1024, 512×512 resolution, and a learning rate that decays exponentially from $1.2e-4$ to $1.2e-6$. Inference uses median compilation ensembling with a cosine noise variance schedule, and, per our scaling experiments, 200 denoising steps with $N = 5$ ensembled samples. Table 3 shows the result: our model matches Marigold’s validation performance on Hypersim and surpasses it on ETH3D, despite training on lower-resolution images with roughly **three orders of magnitude less pre-training data and compute**.

Method	Hypersim		ETH3D		NYUv2		ScanNet		DIODE	
	AbsRel ↓	$\delta 1$ ↑	AbsRel ↓	$\delta 1$ ↑	AbsRel ↓	$\delta 1$ ↑	AbsRel ↓	$\delta 1$ ↑	AbsRel ↓	$\delta 1$ ↑
DiverseDepth	—	—	22.8	69.4	11.7	87.5	10.9	88.2	37.6	63.1
MiDaS	—	—	18.4	75.2	11.1	88.5	12.1	84.6	33.2	71.5
LeReS	—	—	17.1	77.7	9.0	91.6	9.1	91.7	27.1	76.6
Omnidata	—	—	16.6	77.8	7.4	94.5	7.5	93.6	33.9	74.2
HDN	—	—	12.1	83.3	6.9	94.8	8.0	93.9	24.6	78.0
DPT	—	—	7.8	94.6	9.8	90.3	8.2	93.4	18.2	75.8
Marigold	13.5	87.5	6.5	96.0	5.5	96.4	6.4	95.1	30.8	77.3
Ours	13.6	87.6	4.8	97.8	6.8	95.0	7.7	93.7	31.0	77.2

Table 3. **Depth Estimation Performance Comparison on Multiple Datasets.** We achieve state-of-the-art performance on the ETH3D dataset and competitive performance across all other benchmarks. Notably, we closely match the performance of Marigold across all datasets with significantly less training compute.

6.2 Optical Flow Prediction

Optical flow estimation asks for the motion of objects between consecutive video frames, expressed as a dense, pixel-wise displacement field. We carry over essentially the same recipe used for depth estimation: a DiT-XL model trained on FlyingChairs for 40K steps, batch size 1024, 512×512 resolution, and a learning rate decaying exponentially from $1.2e-4$ to $1.2e-6$. Table 4 compares our model against specialized optical flow methods.

Method	FlyingChairs EPE ↓
DeepFlow	3.53
FlowNetS	2.71
FlowNetS+v	2.86
FlowNetS+ft	3.04
FlowNetS+ft+v	3.03
FlowNetC	2.19
FlowNetC+v	2.61
FlowNetC+ft	2.27
FlowNetC+ft+v	2.67
Ours (w/o ensembling)	3.45
Ours (w/ ensembling)	3.08

Table 4. **Optical Flow Comparison with Specialized Techniques.** We evaluate our optical flow model on the FlyingChairs validation set. Our model achieves similar end-point error as specialized methods, including DeepFlow [52] and FlowNet [12]. We train with significantly less data compared to other specialized methods, which use a several optical flow datasets. We generate predictions with and without test-time ensembling.

6.3 Amodal Segmentation

Amodal segmentation asks the model to infer the full shape and extent of an object, including parts hidden from view, which often demands higher-level scene reasoning for cluttered images. We fine-tune a DiT-XL model on the pix2gestalt dataset [32] for 6K steps, using a batch size of 4096, 256×256 resolution, and the same exponentially decaying learning rate from $1.2e-4$ to $1.2e-6$. Table 5 compares our results against other methods.

Method	COCO-A	P2G	MP3D
PCNet	81.35	–	–
PCNet-Sup	82.53	–	–
SAM	67.21	–	–
SD-XL Inpainting	76.52	–	–
pix2gestalt	82.9	88.7	61.5
Ours	82.9	88.6	63.9

Table 5. **Amodal Segmentation Performance (mIOU) Comparison Across Different Datasets.** This table compares mIOU performance across COCO-A, Pix2Gestalt, and MP3D datasets, showing the effectiveness of various methods. Our method is able to achieve competitive performance across all tasks, while training only on Pix2Gestalt.

6.4 One Model for All

Finally, we train a single generalist DiT-XL model across all three tasks at once, using a mixed dataset that combines them. To let the model distinguish between task-specific embeddings during fine-tuning, we replace the standard patch embedding layer with a PatchEmbedRouter module, which routes each VAE embedding to the input convolutional layer appropriate for its task. Training otherwise mirrors our earlier recipes: 512×512 resolution images and a learning rate decaying exponentially from $1.2e-4$ to $1.2e-6$. We then upcycle the fine-tuned DiT-XL checkpoint into a DiT-XL-8E2A model and continue fine-tuning for a further 4K iterations. Fig. 11 shows the resulting predictions, which highlight how well our scaling techniques generalize and transfer across different perception tasks.

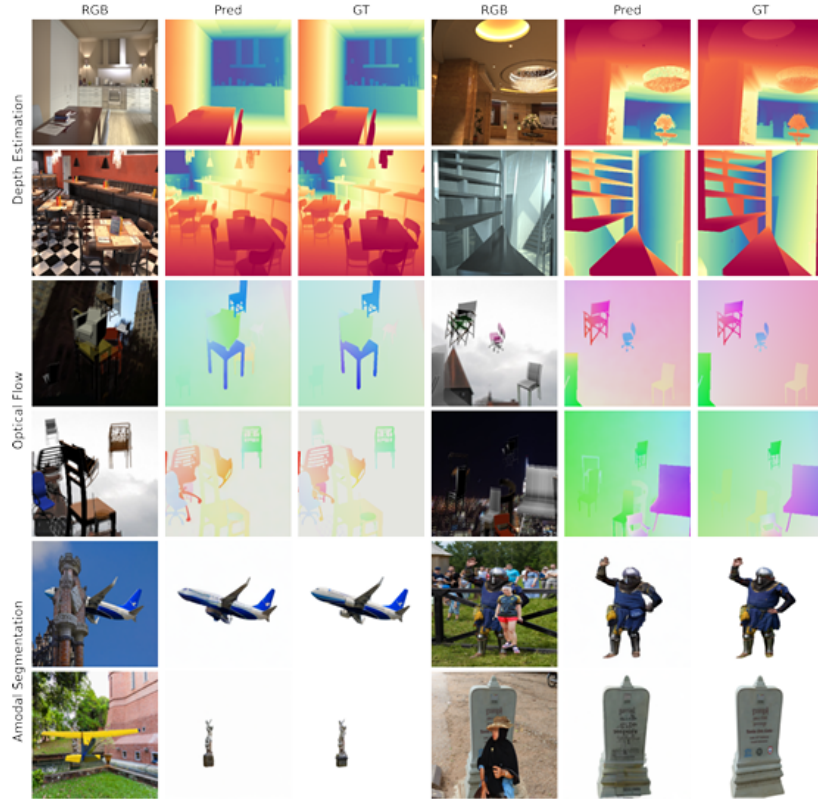


Fig. 11. **Depth Estimation, Optical Flow Estimation, and Amodal Segmentation Examples:** Each row showcases results from our models for different tasks. (a) Depth estimation, with relative scale and shift. (b) Optical flow, with scale and shift. (c) Amodal segmentation, where the model sees an RGB image and segmentation of the occluded object; the task is to predict the amodal image

7 Conclusion

This work examines how diffusion models scale for visual perception tasks. We explore several routes to scaling diffusion training, including larger model size, mixture-of-experts architectures, higher image resolution, and upcycling, and we show that test-time compute can also be scaled efficiently by exploiting diffusion’s iterative structure, yielding significant downstream gains. Across the board, our experiments uncover consistent power laws for both training and inference scaling. We hope these findings encourage further work on scaling both training and test-time compute for iterative generative paradigms like diffusion, particularly for perception tasks.

References

- [1] Tomer Amit, Tal Shaharbany, Eliya Nachmani, and Lior Wolf. Segdiff: Image segmentation with diffusion probabilistic models. *arXiv preprint arXiv:2112.00390*, 2021.
- [2] Dmitry Baranchuk, Ivan Rubachev, Andrey Voynov, Valentin Khulkov, and Artem Babenko. Label efficient semantic segmentation with diffusion models. *arXiv preprint arXiv:2112.03126*, 2021.
- [3] Emmanuel Asiedu Brempong, Simon Kornblith, Ting Chen, Niki Parmar, Matthias Minderer, and Mohammad Norouzi. Denoising pretraining for semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4175–4186, 2022.

- [4] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. Instructpix2pix: Learning to follow image editing instructions. *arXiv preprint arXiv:2211.09800*, 2023.
- [5] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*, 2024.
- [6] Huiwen Chang, Han Zhang, Jarred Barber, AJ Maschinot, Jose Lezama, Lu Jiang, Ming-Hsuan Yang, Kevin Murphy, William T. Freeman, Michael Rubinstein, Yanzhen Li, and Dilip Krishnan. Muse: Text-to-image generation via masked generative transformers. *arXiv preprint arXiv:2301.00704*, 2023.
- [7] Ting Chen, Lala Li, Saurabh Saxena, Geoffrey Hinton, and David J. Fleet. A generalist framework for panoptic segmentation of images and videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 909–919, 2023.
- [8] Yiqun Duan, Xianda Guo, and Zheng Zhu. Diffusiondepth: Diffusion denoising approach for monocular depth estimation. *arXiv preprint arXiv:2303.05021*, 2023.
- [9] Karim El-Refai, Zeeshan Patel, Jonathan Pei, and Tianle Li. Swag: Storytelling with action guidance, 2024.
- [10] Zhengcong Fei, Mingyuan Fan, Changqian Yu, Debang Li, and Junshi Huang. Scaling diffusion transformers to 16 billion parameters. *arXiv preprint arXiv:2407.11633*, 2024.
- [11] Zhengcong Fei, Mingyuan Fan, Changqian Yu, Debang Li, and Junshi Huang. Scaling diffusion transformers to 16 billion parameters. *arXiv preprint*, 2024.
- [12] Philipp Fischer, Alexey Dosovitskiy, Eddy Ilg, Philip Häusser, Caner Hazırbaş, Vladimir Golkov, Patrick van der Smagt, Daniel Cremers, and Thomas Brox. FlowNet: Learning optical flow with convolutional networks. *arXiv preprint arXiv:1504.06852*, 2015.
- [13] Xiao Fu, Wei Yin, Mu Hu, Kaixuan Wang, Yuexin Ma, Ping Tan, Shaojie Shen, Dahua Lin, and Xiaoxiao Long. Geowizard: Unleashing the diffusion priors for 3d geometry estimation from a single image. *arXiv preprint arXiv:2403.12013*, 2024.
- [14] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2014.
- [15] Zhangxuan Gu, Haoxing Chen, and Zhuoer Xu. Diffusioninst: Diffusion model for instance segmentation. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 2730–2734, 2024.
- [16] Wanggui He, Siming Fu, Mushui Liu, Xierui Wang, Wenyi Xiao, Fangxun Shu, Yi Wang, Lei Zhang, Zhelun Yu, Haoyuan Li, Ziwei Huang, LeiLei Gan, and Hao Jiang. Mars: Mixture of auto-regressive models for fine-grained text-to-image synthesis. *arXiv preprint arXiv:2407.07614*, 2024.
- [17] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020.
- [18] Emiel Hoogeboom, Didrik Nielsen, Priyank Jaini, Patrick Forré, and Max Welling. Argmax flows and multinomial diffusion: Learning categorical distributions. *Advances in Neural Information Processing Systems*, 34:12454–12465, 2021.
- [19] Ajay Jain, Ben Mildenhall, Jonathan T. Barron, Pieter Abbeel, and Ben Poole. Zero-shot text-guided object generation with dream fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 867–876, 2022.
- [20] Yuanfeng Ji, Zhe Chen, Enze Xie, Lanqing Hong, Xihui Liu, Zhaoqiang Liu, Tong Lu, Zhenguo Li, and Ping Luo. Ddp: Diffusion model for dense visual prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21741–21752, 2023.
- [21] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing diffusion-based image generators for monocular depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9492–9502, 2024.
- [22] Diederik P. Kingma. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.
- [23] Aran Komatsuzaki, Joan Puigcerver, James Lee-Thorp, Carlos Riquelme Ruiz, Basil Mustafa, Joshua Ainslie, Yi Tay, Mostafa Dehghani, and Neil Houlsby. Sparse upcycling: Training mixture-of-experts from dense checkpoints. *arXiv preprint arXiv:2212.05055*, 2022.
- [24] Aran Komatsuzaki, Joan Puigcerver, James Lee-Thorp, Carlos Riquelme Ruiz, Basil Mustafa, Joshua Ainslie, Yi Tay, Mostafa Dehghani, and Neil Houlsby. Sparse upcycling: Training mixture-of-experts from dense checkpoints. *arXiv preprint arXiv:2212.05055*, 2023.
- [25] Hao Li, Yang Zou, Ying Wang, Orchid Majumder, Yusheng Xie, R. Manmatha, Ashwin Swaminathan, Zhuowen Tu, Stefano Ermon, and Stefano Soatto. On the scalability of diffusion-based text-to-image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9400–9409, 2024.
- [26] Yaron Lipman, Ricky T.Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching: Symmetrizing optimal transport and generative modeling. *arXiv preprint arXiv:2301.13003*, 2023.
- [27] Ruoshi Liu, Rundui Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot one image to 3d object. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9298–9309, 2023.
- [28] Xingchao Liu, Chengyue Gong, and Qiang Liu. Rectified flow: A unified approach for free-form generative models. *arXiv preprint arXiv:2209.07953*, 2022.
- [29] Ao Luo, Xin Li, Fan Yang, Jiangyu Liu, Haoqiang Fan, and Shuaicheng Liu. Flowdiffuser: Advancing optical flow estimation with diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19167–19176, 2024.
- [30] Alex Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. *arXiv preprint arXiv:2102.09672*, 2021.
- [31] OpenAI. Learning to reason with llms. <https://openai.com/index/learning-to-reason-with-llms/>, September 2024.

- [32] Ege Ozguroglu, Ruoshi Liu, Didac Suris, Dian Chen, Achal Dave, Pavel Tokmakov, and Carl Vondrick. pix2gestalt: Amodal segmentation by synthesizing wholes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [33] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4195–4205, 2023.
- [34] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022.
- [35] Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International Conference on Machine Learning*, pages 1530–1538. PMLR, 2015.
- [36] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022.
- [37] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.
- [38] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L. Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in Neural Information Processing Systems*, 35:36479–36494, 2022.
- [39] Saurabh Saxena, Abhishek Kar, Mohammad Norouzi, and David J. Fleet. Monocular depth estimation using diffusion models. *arXiv preprint arXiv:2302.14816*, 2023.
- [40] Saurabh Saxena, Charles Herrmann, Junhwa Hur, Abhishek Kar, Mohammad Norouzi, Deqing Sun, and David J. Fleet. The surprising effectiveness of diffusion models for optical flow and monocular depth estimation. *Advances in Neural Information Processing Systems*, 36, 2024.
- [41] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziars, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- [42] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- [43] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International Conference on Machine Learning*, pages 2256–2265. PMLR, 2015.
- [44] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.
- [45] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. *arXiv preprint arXiv:2303.01469*, 2023.
- [46] Haotian Sun, Tao Lei, Bowen Zhang, Yanghao Li, Haoshuo Huang, Ruoming Pang, Bo Dai, and Nan Du. Ec-dit: Scaling diffusion transformers with adaptive expert-choice routing. *arXiv preprint arXiv:2410.02098*, 2024.
- [47] Haoru Tan, Sitong Wu, and Jimin Pi. Semantic diffusion network for semantic segmentation. *Advances in Neural Information Processing Systems*, 35:8702–8716, 2022.
- [48] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shrutie Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [49] Aäron van den Oord, Nal Kalchbrenner, and Koray Kavukcuoglu. Pixel recurrent neural networks. In *International Conference on Machine Learning*, pages 1747–1756. PMLR, 2016.
- [50] Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A. Yeh, and Greg Shakhnarovich. Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12619–12629, 2023.
- [51] Daniel Watson, William Chan, Ricardo Martin-Brualla, Jonathan Ho, Andrea Tagliasacchi, and Mohammad Norouzi. Novel view synthesis with diffusion models. *arXiv preprint arXiv:2210.04628*, 2022.
- [52] Philippe Weinzaepfel, Jerome Revaud, Zaid Harchaoui, and Cordelia Schmid. Deepflow: Large displacement optical flow with deep matching. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 1385–1392, 2013.
- [53] Julia Wolleb, Robin Sandkühler, Florentin Bieder, Philippe Valmaggia, and Philippe C. Cattin. Diffusion models for implicit image segmentation ensembles. In *International Conference on Medical Imaging with Deep Learning*, pages 1336–1348. PMLR, 2022.
- [54] Greg Yang, Edward J. Hu, Igor Babuschkin, Szymon Sidor, Xiaodong Liu, David Farhi, Nick Ryder, Jakub Pachocki, Weizhu Chen, and Jianfeng Gao. Tensor programs v: Tuning large neural networks via zero-shot hyperparameter transfer. *arXiv preprint arXiv:2203.03466*, 2022.
- [55] Jiahui Yu, Yuanzhong Xu, Jing Yu Koh, Thang Luong, Gunjan Baid, Zirui Wang, Vijay Vasudevan, Alexander Ku, Yinfei Yang, Burcu Karagol Ayan, et al. Scaling autoregressive models for content-rich text-to-image generation. *arXiv preprint arXiv:2206.10789*, 2022.
- [56] Wenliang Zhao, Yongming Rao, Zuyan Liu, Benlin Liu, Jie Zhou, and Jiwen Lu. Unleashing text-to-image diffusion models for visual perception. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5729–5739, 2023.